

Desktop PC Teardown, Component Identification & Reassembly

Hands-On Systems Hardware Diagnostic Lab

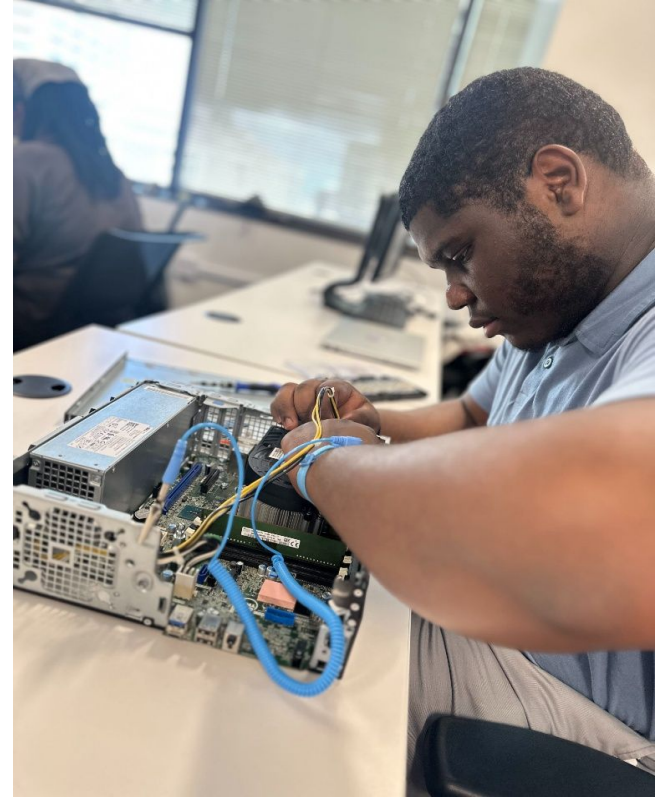
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CompTIA A+ Practical Diagnostics

ESD Safety & Workspace Preparation

ESD Compliance: Wore anti-static grounding wrist straps connected to a common ground point to prevent electrostatic damage to sensitive ICs.

Workplace Organization: Prepared a clean, non-conductive surface to organize screws, brackets, and modular cables systematically.

Pre-Teardown Functional Check: Verified initial power delivery and POST status prior to chassis opening.

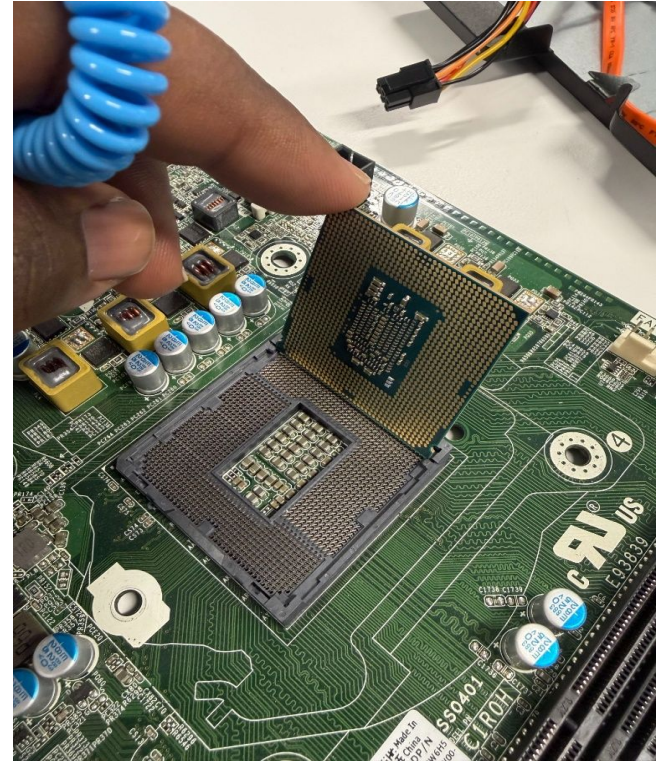


Processor Removal & Socket Pin Inspection

LGA Socket Architecture: Released retention arm to safely inspect the Land Grid Array (LGA) socket for bent pins or debris.

Thermal Interface Material (TIM): Cleaned legacy thermal compound using isopropyl alcohol prior to reseating.

Alignment & Zero-Force Seating: Oriented Pin 1 (gold triangle indicator) with socket notch guides to prevent pin destruction during seating.



System Board Extraction & Bus Interfaces

Chassis Removal: Unfastened standoffs and detached front panel header leads to extract the Dell OEM system board.

Expansion Slot Identification: Documented PCI Express x16, PCIe x1 slots, SATA interfaces, and DIMM slot configurations.

Voltage Regulation Modules (VRMs): Mapped VRM inductors and solid capacitors responsible for CPU power delivery.



Component Integration & Power Architecture

M.2 NVMe Storage Integration: Installed PCIe M.2 SSD using retention screw anchor for high-speed boot drive storage.

Dual-Channel RAM Configuration: Populated DDR4 memory modules into primary channels for optimal performance.

Proprietary PSU Cabling: Mapped OEM 6-pin power connector distribution rails and CPU 4-pin ATX power lines.



Final System Reassembly & Diagnostics

Cable Management: Secured internal wire runs away from CPU cooling fan and heatsinks to maintain optimal airflow.

Power-On Self-Test (POST): Reconnected power and peripheral I/O to confirm normal POST sequence and BIOS hardware detection.

Lab Outcome: Successfully disassembled, documented, and rebuilt system without hardware faults or ESD incidents.

